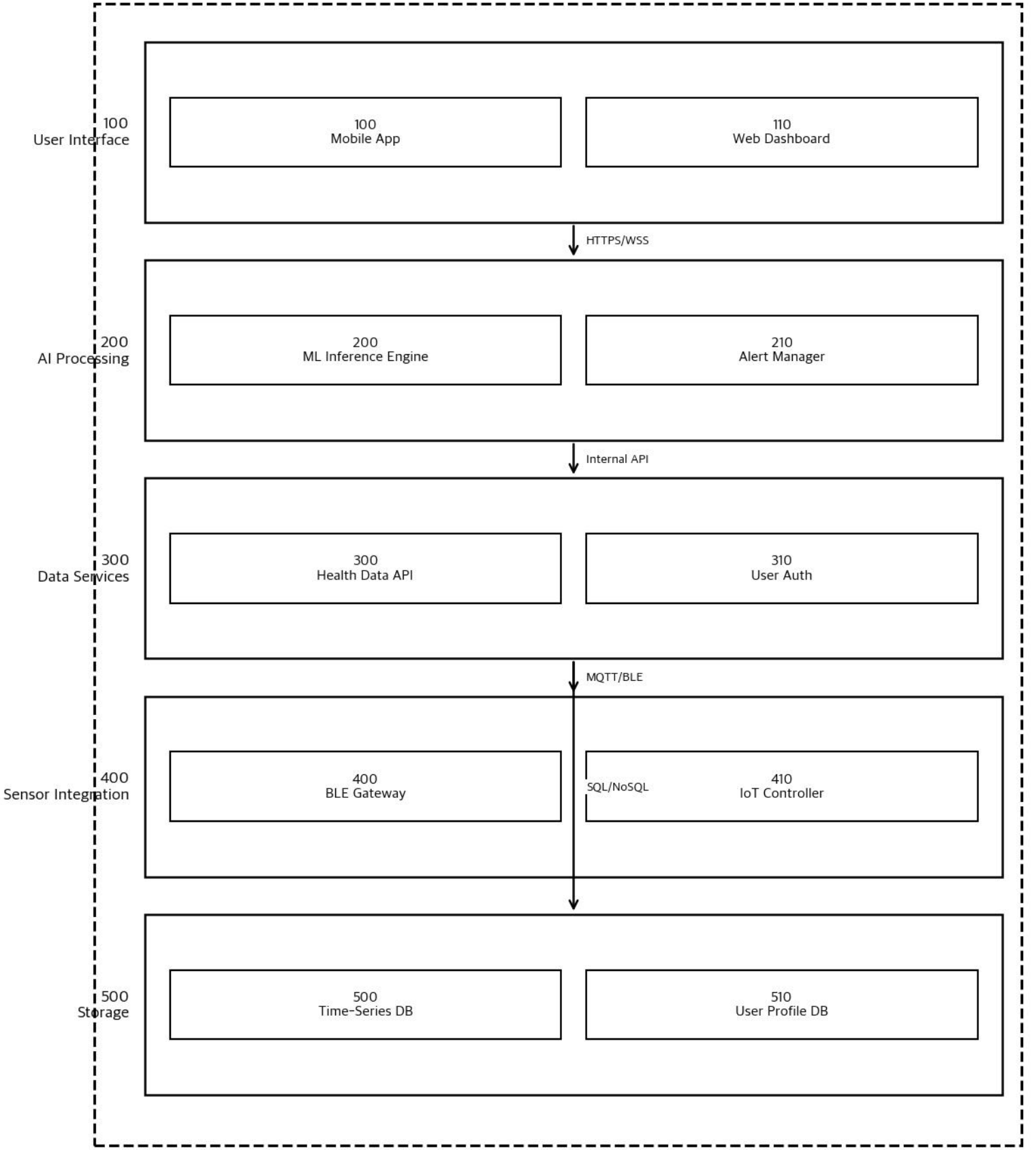
100
User Interface
100
Mobile App
110
Web Dashboard
HTTPS/WSS
200
AI Processing
200
ML Inference Engine
210
Alert Manager
Internal API
300
Data Services
300
Health Data API
310
User Auth
MQTT/BLE
400
Sensor Integration
400
BLE Gateway
SQL/NoSQL
410
IoT Controller
500
Storage
500
Time-Series DB
510
User Profile DB

FIG. 1

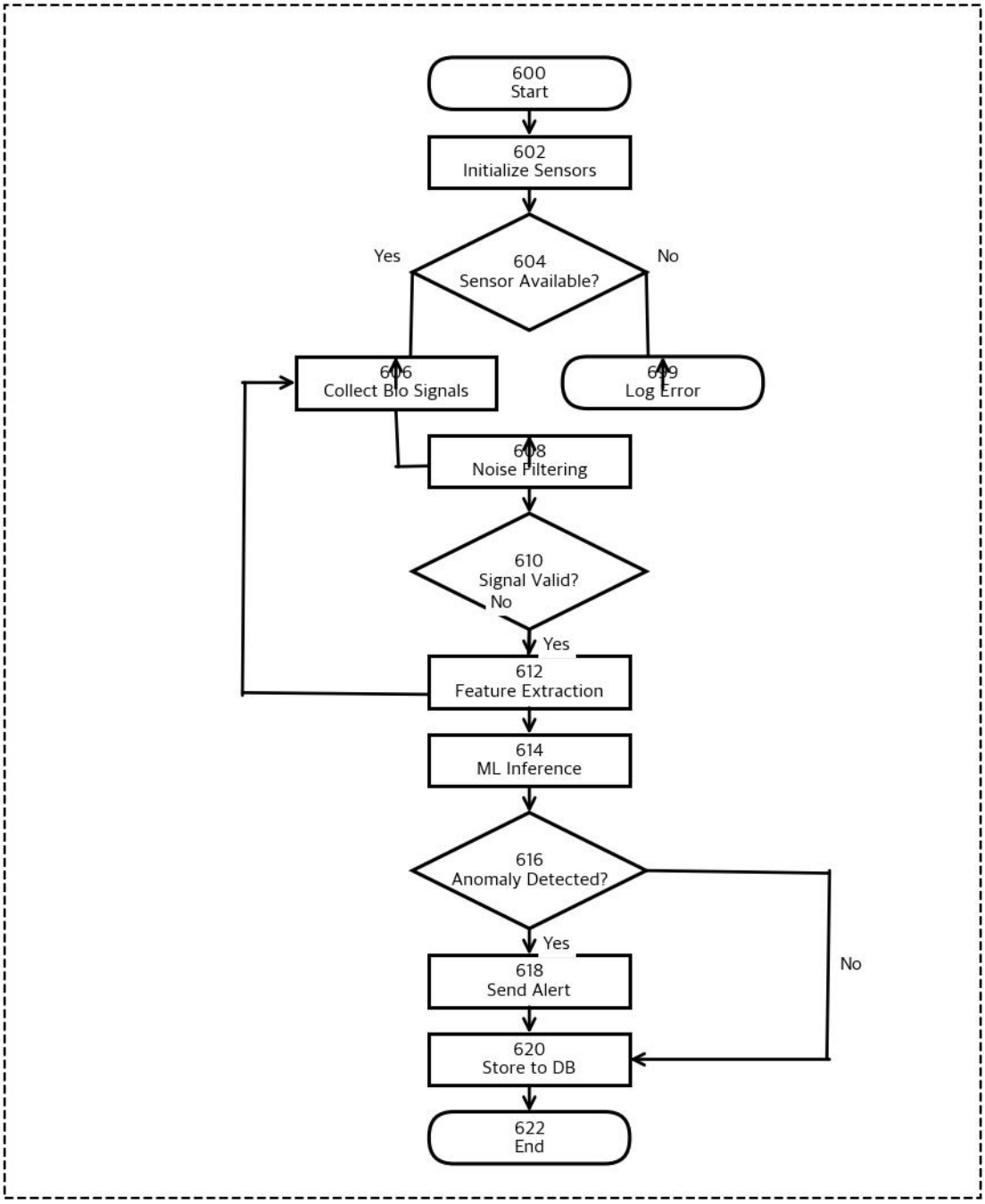
600
Start
602
Initialize Sensors
604
Sensor Available?
Yes
No
606
Collect Bio Signals
609
Log Error
608
Noise Filtering
610
Signal Valid?
No
Yes
612
Feature Extraction
614
ML Inference
616
Anomaly Detected?
Yes
No
618
Send Alert
620
Store to DB
622
End

FIG. 2

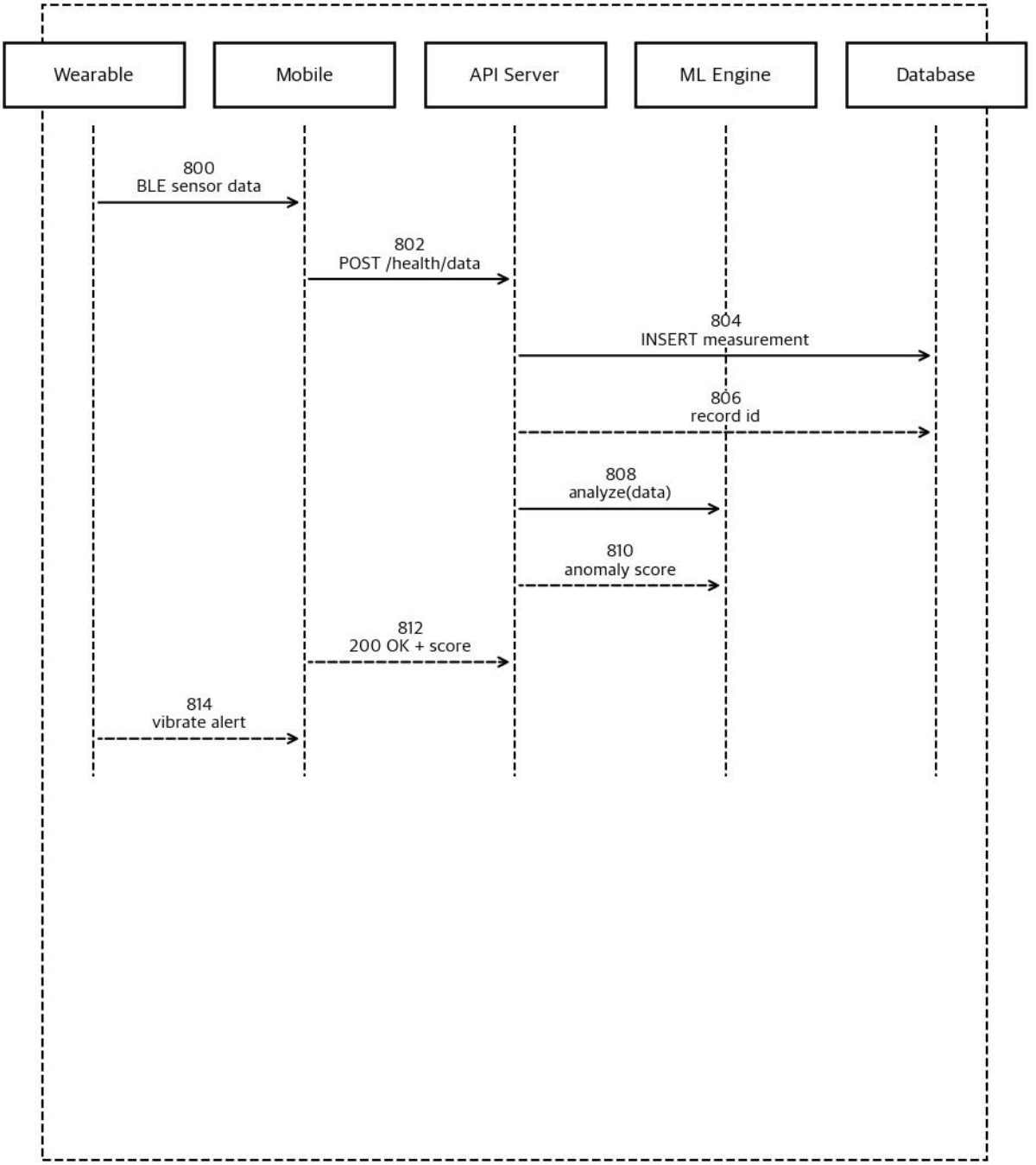
Wearable
Mobile
API Server
ML Engine
Database
800
BLE sensor data
802
POST /health/data
804
INSERT measurement
806
record id
808
analyze(data)
810
anomaly score
812
200 OK + score
814
vibrate alert

FIG. 3

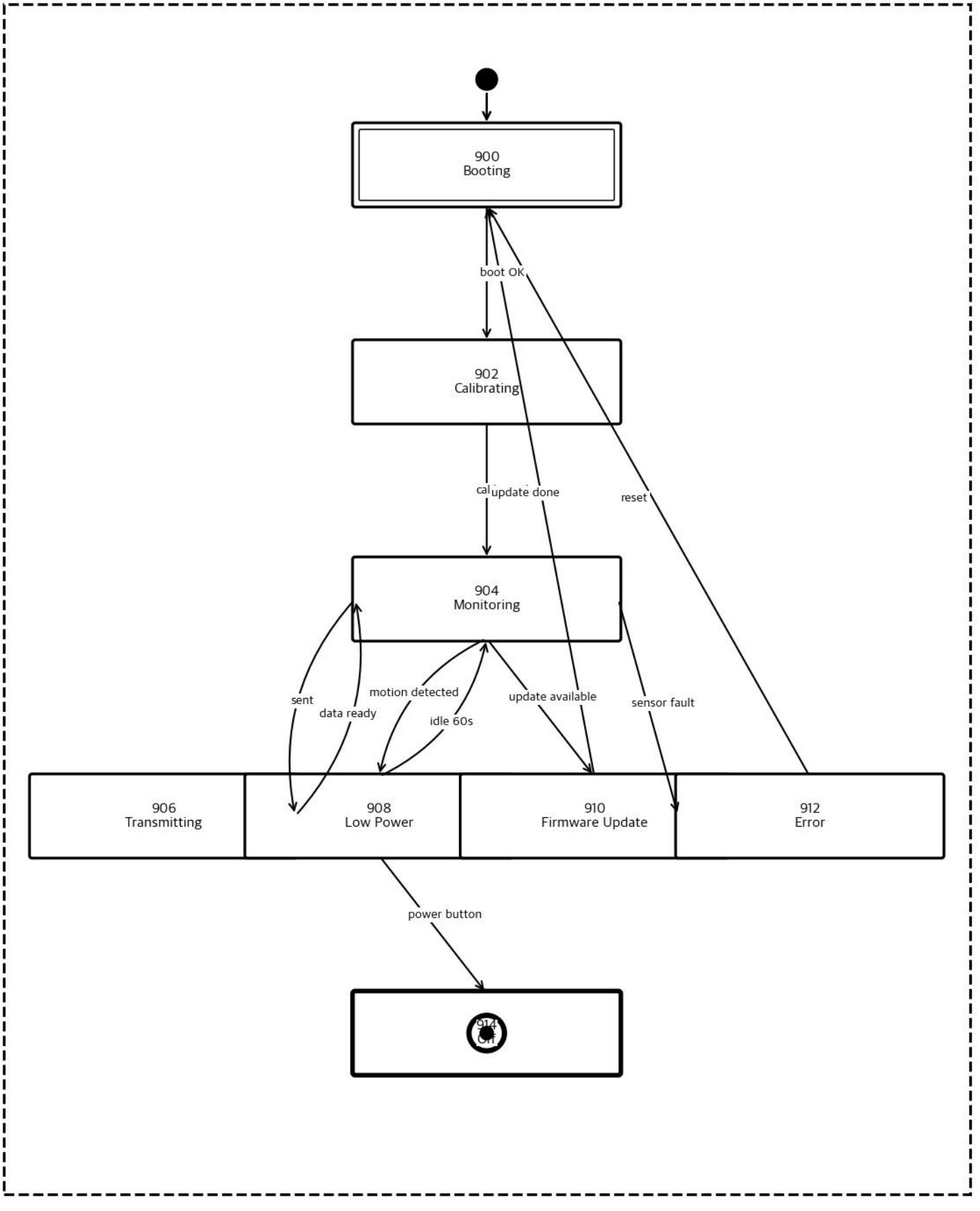
900
Booting
boot OK
902
Calibrating
cal
update done
reset
904
Monitoring
sent
data ready
motion detected
idle 60s
update available
sensor fault
906
Transmitting
908
Low Power
910
Firmware Update
912
Error
power button
914
Off

FIG. 4

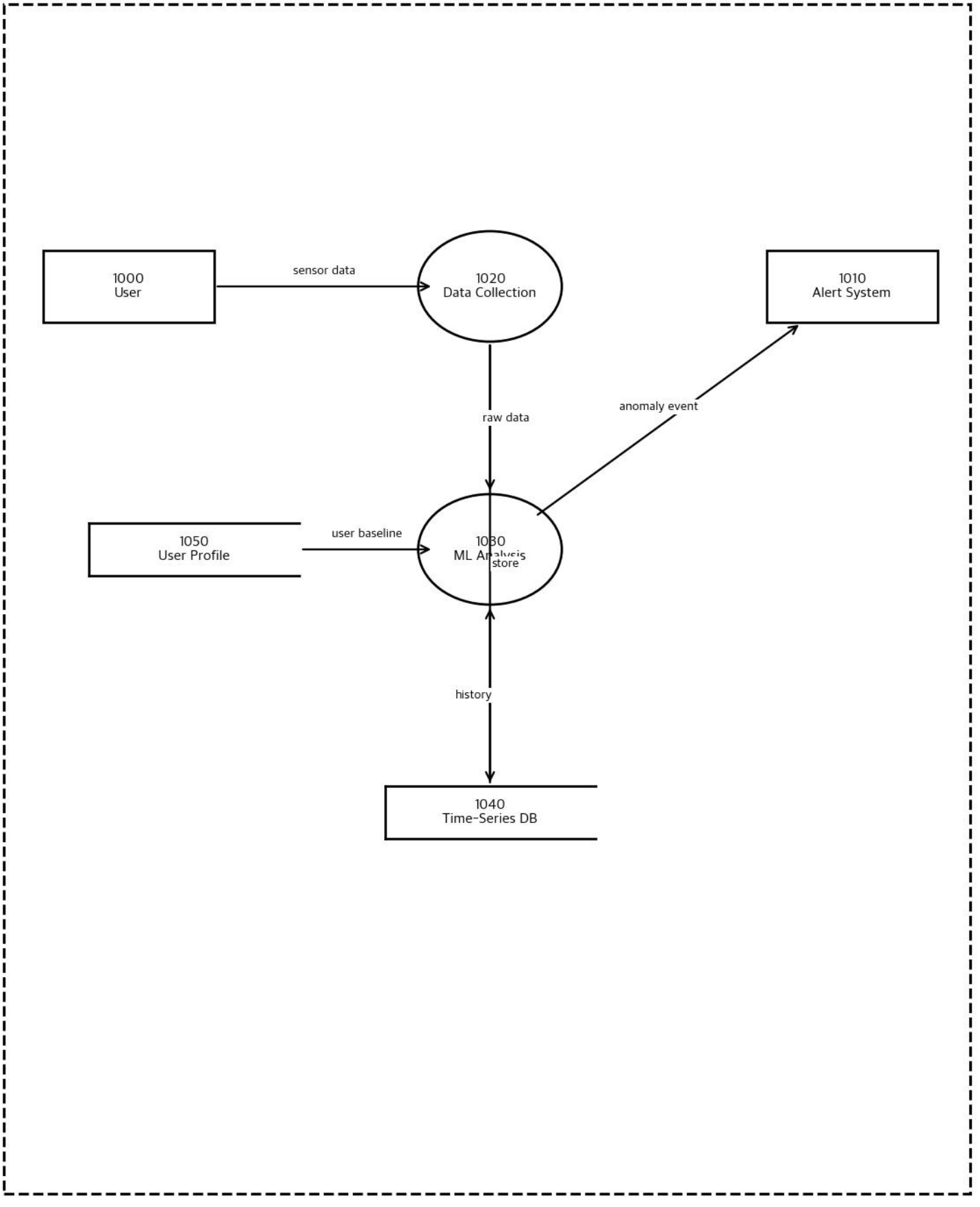
1000
User
sensor data
1020
Data Collection
1010
Alert System
raw data
anomaly event
1050
User Profile
user baseline
1030
ML Analysis
store
history
1040
Time-Series DB

FIG. 5

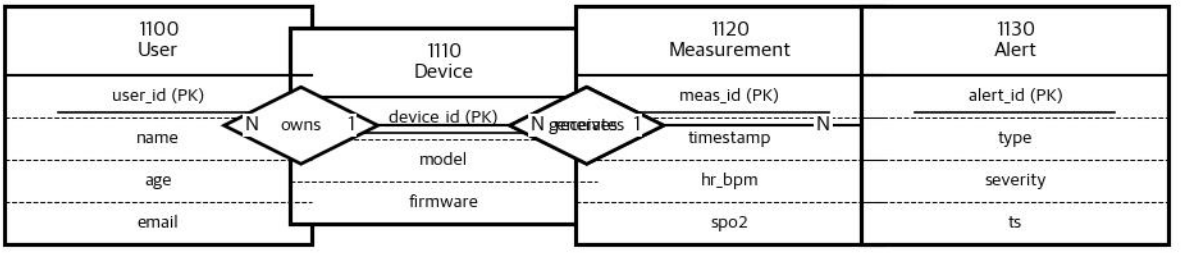
1100
User
user_id (PK)
name
age
email
N
owns
1
1110
Device
device_id (PK)
model
firmware
N
generates
1
1120
Measurement
meas_id (PK)
timestamp
hr_bpm
spo2
N
1130
Alert
alert_id (PK)
type
severity
ts

FIG. 6

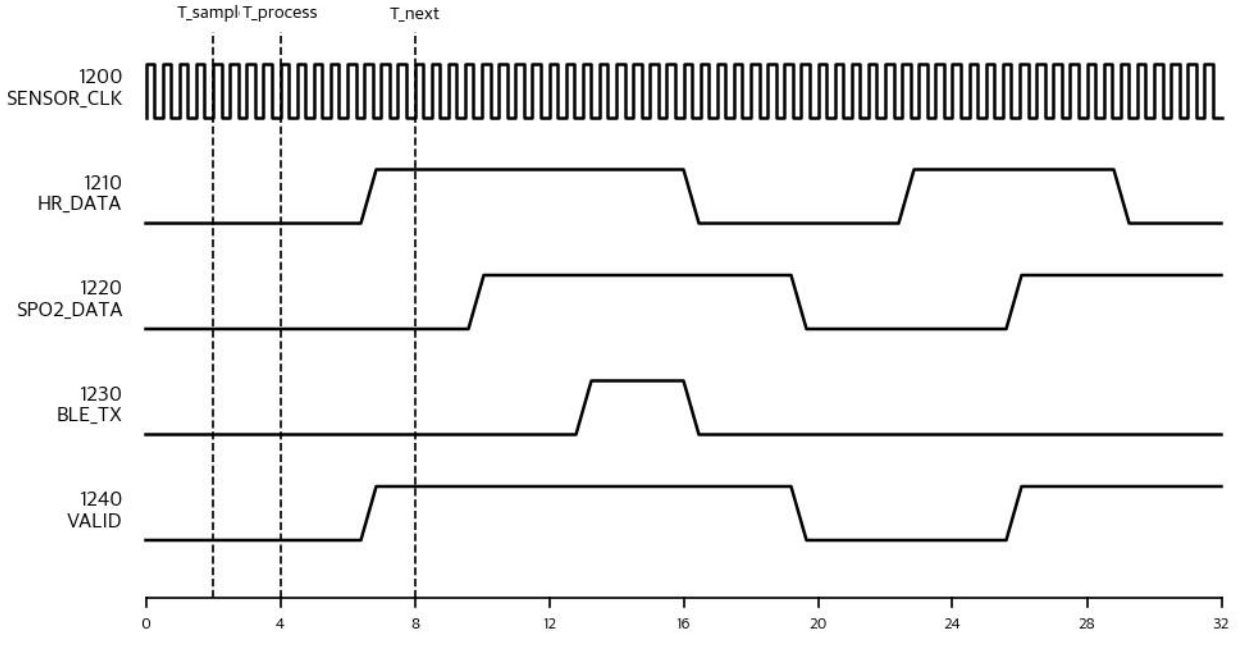
T_sampl
T_process
T_next
1200
SENSOR_CLK
1210
HR_DATA
1220
SPO2_DATA
1230
BLE_TX
1240
VALID
0
4
8
12
16
20
24
28
32

FIG. 7